

Vikram Watanabe

Embedded Systems & IoT Firmware Engineer • 12 Years Experience • MERIQ Index: 81% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, Python, I2C/SPI/UART, FreeRTOS, PCB Debugging, MQTT, ARM Cortex

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 46%.
- Led a squad of 4 engineers delivering scalable microservices processing 7M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 53%.

Associate Embedded Systems & IoT Firmware Engineer — *Tesla* (2014 - 2023)

- Designed and maintained Python data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$14k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [Embedded Linux, Python, I2C/SPI/UART, FreeRTOS]

Engineered end-to-end production solution leveraging Embedded Linux and Python with full test automation.

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [I2C/SPI/UART, FreeRTOS, PCB Debugging, MQTT]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

UC Berkeley — 2010 - 2014 (Score: 7.94 / 10.0)